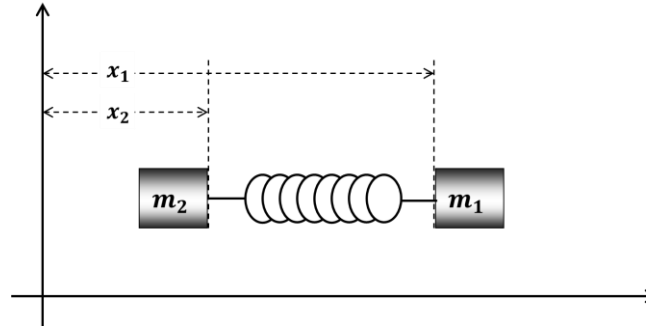


Two Body Oscillations:

Let us consider two bodies m_1 and m_2 are connected by a massless spring of force constant K as shown in Figure. Assume the system is free to oscillate on a frictionless horizontal surface. Here we can locate the ends of the spring by coordinate $x_1(t)$ and $x_2(t)$ as shown in the figure.



Hence, the length of the spring at any instant is $x_1 - x_2$. If its normal, unstressed length is l , then the change in length of the spring, $x(t)$ is given by:

$$x = (x_1 - x_2) - l \quad (1)$$

We see that, if x is positive, the spring is stretched, if $x = 0$, the spring has its normal length, and if x is negative, it is compressed.

We can write the differential equation for two masses m_1 and m_2 respectively:

$$m_1 \frac{d^2 x_1}{dt^2} = -Kx \quad (2)$$

$$\text{and } m_2 \frac{d^2 x_2}{dt^2} = -Kx \quad (3)$$

Let us now multiply Eqⁿ(1) by m_2 and Eqⁿ(2) by m_1 and subtract. We obtain:

$$\begin{aligned} m_1 m_2 \frac{d^2 x_1}{dt^2} - m_1 m_2 \frac{d^2 x_2}{dt^2} &= -m_2 Kx - m_1 Kx \\ \Leftrightarrow m_1 m_2 \frac{d^2}{dt^2} (x_1 - x_2) &= -(m_1 + m_2) Kx \\ \Leftrightarrow \frac{m_1 m_2}{m_1 + m_2} \frac{d^2}{dt^2} (x_1 - x_2) &= -Kx \end{aligned} \quad (4)$$

Let us call the quantity $\frac{m_1 m_2}{m_1 + m_2}$, which has the dimension of mass, the reduced mass of the system and give it the symbol μ , that is –

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \quad (5)$$

Because l is constant, $d^2(x_1 - x_2)/dt^2 = d^2x/dt^2$ and Eqⁿ(4) now can be written as:

$$\begin{aligned} \mu \frac{d^2 x}{dt^2} &= -Kx \\ \Leftrightarrow \frac{d^2 x}{dt^2} + \frac{K}{\mu} x &= 0 \end{aligned}$$

Finally, $\frac{d^2x}{dt^2} + \omega^2 x = 0$

This is an identical form of the equation for the single body oscillation. So, we can conclude that, it is possible to convert two body system into the single body oscillations, Where –

- a) The mass of the system, $\mu = \frac{m_1 m_2}{m_1 + m_2}$, is always smaller than m_1 or m_2 .
- b) Time Period, $T = 2\pi \sqrt{\frac{\mu}{K}}$
- c) Frequency, $n = \frac{1}{2\pi} \sqrt{\frac{K}{\mu}}$

Problem-1: If two bodies of masses $5g$ and $10g$ are connected by a massless spring of force constant 1.5 dynes/cm , then calculate the effective mass and time period of oscillation.

Solution: Given that, $m_1 = 5g$

$$m_2 = 10g$$

$$K = 1.5 \text{ dynes/cm}$$

$$\mu = ? \text{ and } T = ?$$

We have, $\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{5 \times 10}{5 + 10} = 3.33g$ (Ans.)

Again, $T = 2\pi \sqrt{\frac{\mu}{K}} = 2\pi \sqrt{\frac{3.33}{1.5}} = 9.366 \text{ sec}$ (Ans.)

Problem-2: Two bodies are connected by a massless spring of force constant 1.5 dynes/cm . If the reduced mass of the system be $4g$ and one of the bodies having mass of $3g$, then calculate the time period of the oscillation and the mass of other body connected to the system.

Solution: Given that, $K = \frac{1.5 \text{ dynes}}{\text{cm}}$

$$\mu = 3g$$

$$m_1 = 4g$$

$$T = ? \text{ and } m_2 = ?$$

We have, $T = 2\pi \sqrt{\frac{\mu}{K}} = 2\pi \sqrt{\frac{3}{1.5}} = 8.88 \text{ sec}$ (Ans.)

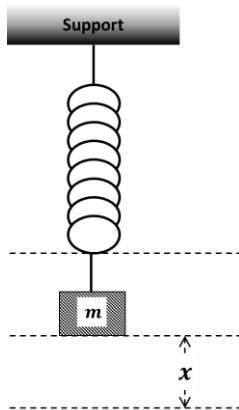
Again, $\mu = \frac{m_1 m_2}{m_1 + m_2} \Leftrightarrow 3 = \frac{4m_2}{4 + m_2} \Leftrightarrow 12 + 3m_2 = 4m_2$

$$\Leftrightarrow 4m_2 - 3m_2 = 12$$

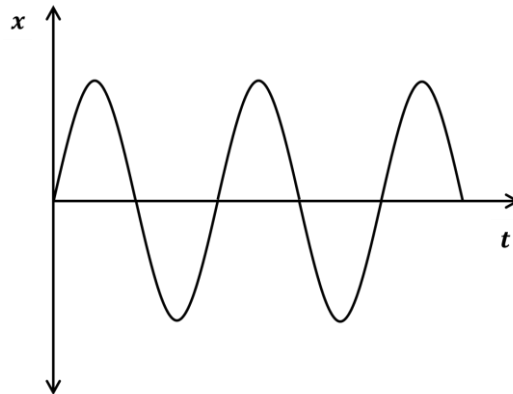
$$m_2 = 12g$$
 (Ans.)

Undamped Oscillation:

When the motion of an oscillator is unaffected by any sort of external forces, then it will oscillate continually with constant amplitude and in that case the total energy of the oscillating system remain same, then the motion of that oscillator is called undamped oscillations or free vibrations.



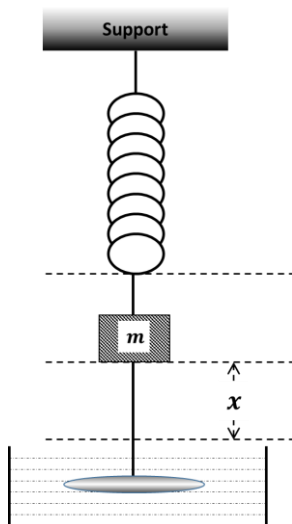
Block Spring Oscillator



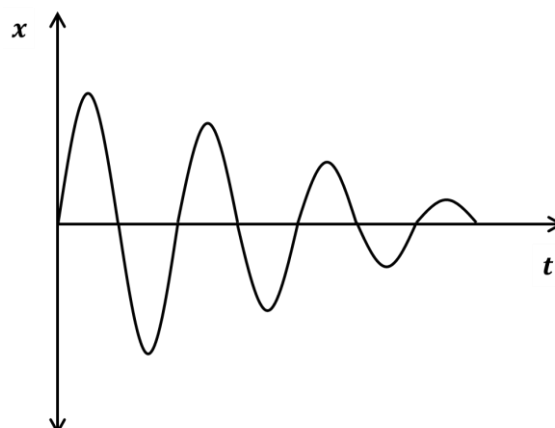
Waveform of Constant Amplitude

Damped Oscillation:

When an external force reduces the motion of an oscillator, the oscillator and its motion are said to be damped. An idealized example of a damped oscillator is shown in figure, where a block with mass m oscillates vertically on a spring with spring constant K . From the block, a rod extends to a vane (both assumed massless) that is submerged in a liquid. As the vane moves up and down, the liquid exerts an inhibiting drag force on it and thus on the entire oscillating system. With time, the mechanical energy is transformed to thermal energy of the liquid and vane.



Block Spring Oscillator



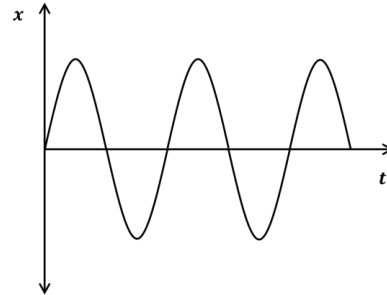
Waveform of Decaying Amplitude

Forced Oscillations:

We came to know that the time period of a body executing simple harmonic motion depends on the dimension of the body and its elastic properties. The oscillations of such a body die out with time due to dissipation energy. If some external periodic force is continually applied on the body, the body continues to oscillate under the influence of such external forces. Such oscillations of the body are called **Forced Oscillations**.



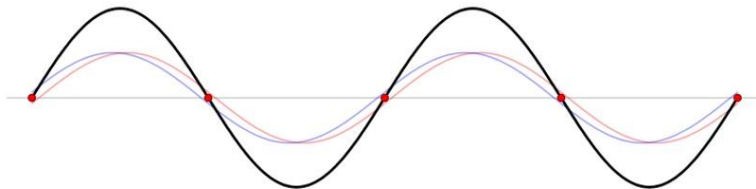
Example of Forced Oscillation



Waveform of Forced Oscillation

Resonance:

The resonance is a special case of forced vibrations. If the frequency and phase of driving force is same as that of the free vibrations. Amplitude of the vibrations keep increasing in magnitude continuously. This is called **resonance**.

**Lissajous Figures:**

When a particle is influenced simultaneously by two simple harmonic motion at right angle to each other, the resultant motion of the particle traces a curve. These curves are called Lissajous' Figures.

The shape of the curve depends on the followings:

- The time period
- Difference in Phase
- Amplitude of constituent oscillations.

